

Exercice 8 : (Intégrale de Wallis)

Soit $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ et $J_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$

- 1) Montrer que $I_n = J_n$.
- 2) Établir une relation de récurrence entre I_n et I_{n+2} .
- 3) En déduire I_{2p} et I_{2p+1} .
- 4) Montrer que $(n+1)I_{n+1}I_n = \frac{\pi}{2}$ pour tout entier n .
- 5) Montrer que $\lim_{n \rightarrow +\infty} \frac{I_{n+1}}{I_n} = 1$. En déduire qu'au voisinage de l'infini $I_n \sim \sqrt{\frac{\pi}{2n}}$

$$I_n = \int_0^{\pi/2} \sin^n x \, dx$$

$$J_n = \int_0^{\pi/2} \cos^n x \, dx$$

1°/ M. Q. $I_n = J_n$

$$I_n = \int_0^{\pi/2} \sin^n x \, dx$$

$$\cos(\pi/2 - x) = \sin(x)$$

$$\sin(\pi/2 - x) = \cos(x)$$

On pose $t = \pi/2 - x \Rightarrow \frac{dt}{dx} = -1 \Rightarrow dx = -dt$

$$\begin{cases} x = \frac{\pi}{2} \\ x = 0 \end{cases} \Rightarrow \begin{cases} t = 0 \\ t = \pi/2 \end{cases}$$

$$\begin{aligned} \text{donc } I_n &= \int_{\pi/2}^0 \sin^n\left(\frac{\pi}{2} - t\right) (-dt) = \int_{\pi/2}^0 \cos^n t \cdot -dt \\ &= - \int_{\pi/2}^0 \cos^n t \, dt = \int_0^{\pi/2} \cos^n t \, dt = J_n \end{aligned}$$

2°/ $I_{n+2} = \int_0^{\pi/2} \sin^{n+2}(t) \, dt = \int_0^{\pi/2} \sin t \cdot \sin^{n+1}(t) \, dt$

$$\begin{cases} u = \sin^{n+1}(t) \\ u' = \sin^n(t) \end{cases} \Rightarrow \begin{cases} u' = (n+1) \sin^n \cos t \\ v = -\cos(t) \end{cases}$$

$$\begin{aligned} \text{donc } I_{n+2} &= \left[-\cos(t) \cdot \sin^{n+2}(t) \right]_0^{\pi/2} + \int_0^{\pi/2} (n+1) \sin^n \cos^2 t \, dt \\ &= (0 - 0) + (n+1) \int_0^{\pi/2} \sin^n t (1 - \sin^2 t) \, dt \end{aligned}$$

$$= (n+1) \left[\int_0^{\pi/2} \sin^n t \, dt - \int_0^{\pi/2} \sin^{n+2}(t) \, dt \right]$$

$$I_{n+2} = n+1 \left[I_n - I_{n+2} \right]$$

$$\Rightarrow I_{n+2} = (n+1) \cdot I_n - (n+1) I_{n+2}$$

$$\Rightarrow I_{n+2} + (n+1) I_{n+2} = (n+1) I_n$$

$$\Rightarrow I_{n+2} = \frac{n+1}{n+2} \cdot I_n$$

So // $n = 2p$ On a: $I_{2p+2} = \frac{2p+1}{2p+2} I_{2p}$

$$\Rightarrow I_{2(p+1)} = \frac{2p+1}{2p+2} \cdot I_{2p}$$

On pose $I_{2p} = U_p$

donc $U_{p+1} = \frac{2p+1}{2p+2} \cdot U_p$

$$\Rightarrow \frac{U_{k+1}}{U_k} = \frac{2k+1}{2k+2}$$

$$\Rightarrow \prod_{k=0}^{p-1} \frac{U_{k+1}}{U_k} = \prod_{k=0}^{p-1} \frac{2k+1}{2k+2}$$

$$\begin{aligned} 2(p-1)+1 &= 2p-1 \\ 2(p-1)+2 &= 2p \end{aligned}$$

$$\Rightarrow \frac{U_p}{U_0} = \frac{1 \times 3 \times 5 \times 7 \times \dots \times 2p-1}{2 \times 4 \times 6 \times 8 \times \dots \times 2p}$$

$$= \frac{1 \times 2 \times 3 \times 4 \times 5 \times \dots \times 2p \times 2p-1}{(2 \times 4 \times 6 \times 8 \times \dots \times 2p) (2 \times 4 \times 6 \times 8 \times \dots \times 2p)}$$

$$= \frac{(2p)!}{(2 \times 4 \times 6 \times 8 \times \dots \times 2p)^2}$$

$$= \frac{(2p)!}{2 \times 1 \times 2 \times 2 \times 3 \times \dots \times 2 \times p}$$

$$\frac{U_p}{U_0} = \frac{(2p)!}{2^p p!}$$

$$\Rightarrow \frac{U_{k+1}}{U_k} = \dots \dots \dots$$

$$\frac{U_1}{U_0} = \dots \dots \dots$$

$$\frac{U_2}{U_0} = \dots$$

$$\Rightarrow \frac{U_1}{U_0} \times \frac{U_2}{U_1} \times \frac{U_3}{U_2} \times \dots \times \frac{U_p}{U_{p-1}}$$

$$= \frac{U_p}{U_0} = \frac{(2p)!}{2^p p!}$$

$$\frac{U_p}{U_{p-1}} = \dots$$

4// Mq $(n+1) \cdot I_n \cdot I_{n+1} = \frac{\pi}{2} \quad \forall n \in \mathbb{N}$

on pose $V_n = (n+1) \cdot I_n \cdot I_{n+1}$

Mq V_n est constante c-à-d : $V_{n+1} = V_n$

$$V_{n+1} - V_n = (n+2) \cdot I_{n+1} \cdot I_{n+2} - (n+1) \cdot I_n \cdot I_{n+1}$$

$$= \cancel{(n+1)} I_{n+1} \cdot \frac{n+2}{n+1} \cdot I_n - (n+1) \cdot I_n \cdot I_{n+1}$$

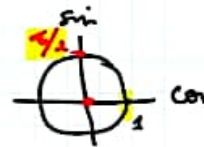
$$V_{n+1} - V_n = 0$$

donc V_n est cte.

donc $\forall n \in \mathbb{N} \quad V_n = V_0 = I_0 \cdot I_1$

$$I_0 = \int_0^{\pi/2} \sin^0(t) dt = \int_0^{\pi/2} 1 \cdot dt = \frac{\pi}{2}$$

$$I_1 = \int_0^{\pi/2} \sin^1(t) dt = \left[-\cos(t) \right]_0^{\pi/2} = -(0 - 1) = 1$$



donc $\forall n \in \mathbb{N} \quad V_n = I_0 \cdot I_1$

$$\Rightarrow V_n = \frac{\pi}{2} \cdot 1 = \frac{\pi}{2}$$

5// Mq $\lim_{n \rightarrow +\infty} \frac{I_{n+1}}{I_n} = 1$

On a : $v_{2p} = \frac{I_{2p+1}}{I_{2p}} = \frac{\frac{(2p)!}{2^p p!}}{\frac{(2p)!}{2^p p!}} \xrightarrow{p \rightarrow +\infty} 1$

$v_{2p+1} = \frac{I_{2p+2}}{I_{2p+1}} \xrightarrow{p \rightarrow +\infty} 1$

$\Rightarrow \lim_{n \rightarrow +\infty} v_n = 1$

• En déduire que $I_n \sim \sqrt{\frac{\pi}{2n}} \quad n+1 \sim n$

On a : $\lim_{n \rightarrow +\infty} \frac{I_{n+1}}{I_n} = 1 \Rightarrow I_{n+1} \sim I_n$

done $(n+1) I_n \cdot I_{n+1} \sim I_n \cdot n I_n$

for $n+1 \rightarrow \infty$

done $\frac{\pi}{2} \sim n \cdot I_n^2$

$$\Rightarrow I_n^2 \sim \frac{\pi}{2n}$$

$$\Rightarrow I_n \sim \sqrt{\frac{\pi}{2n}}$$

lyas?
math,